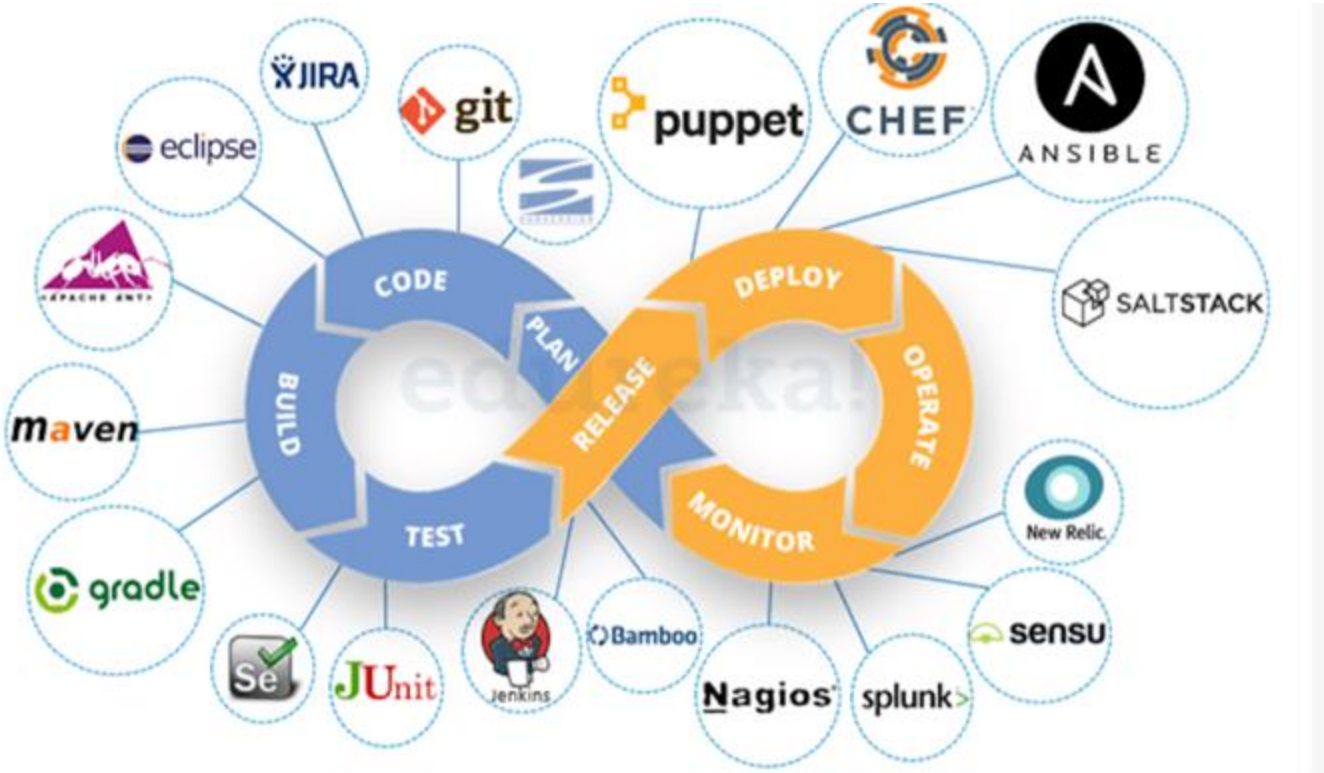
Architecture basics

Architecture basics

- How a set of software and hardware technology standards interact to form a computer system or platform.
- **All architecture executes three steps:**
  - Fetch.
  - Decode.
  - Execute.
- **Three categories:**
  - **System Design:** hardware.
  - **Instruction Set Architecture (ISA):** embedded programming code.
  - **Microarchitecture:** data paths, processing and storage.



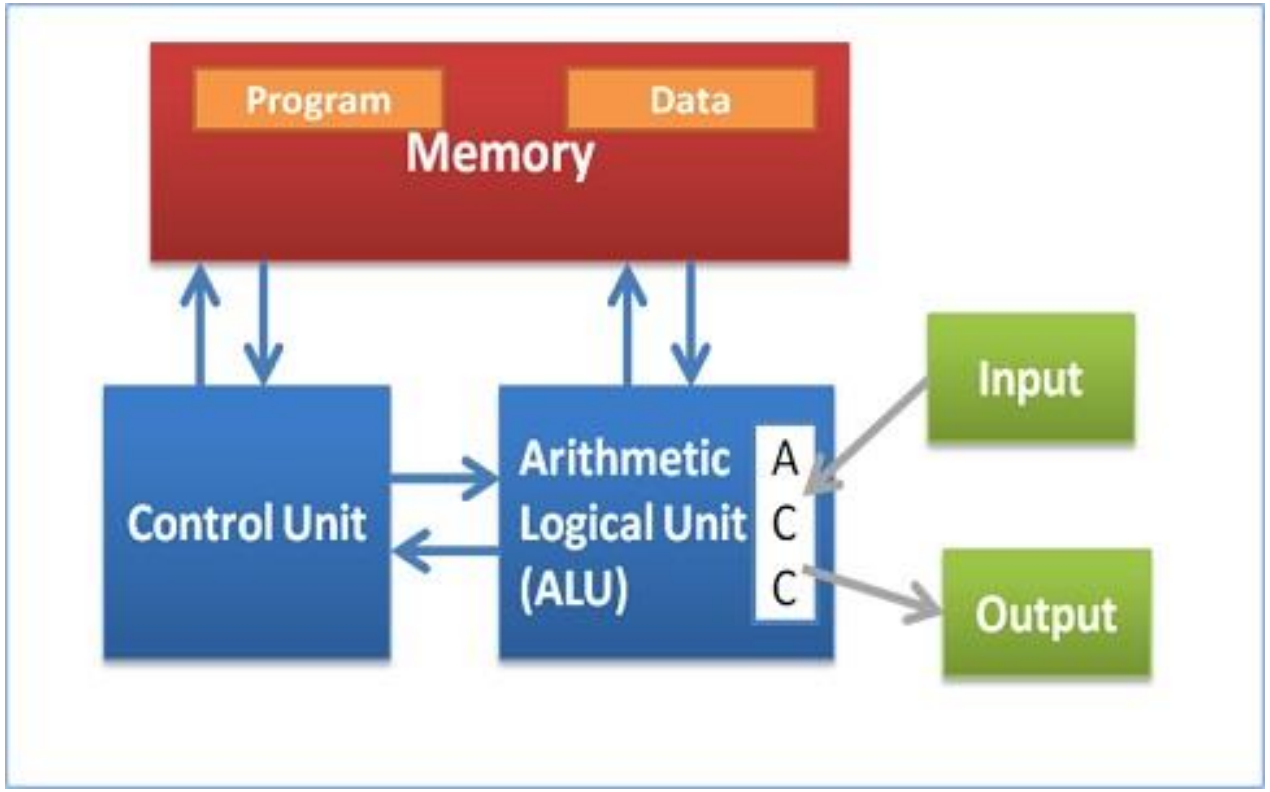
Source: https://medium.com/@AnnFelix/best-intermediate-devops-project-ideas-9d069bf78827

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Architecture models (1/3)

- Store program in memory prior to execution.
- Bottleneck: uses serial processing rather than parallel.

Von Neuman architecture



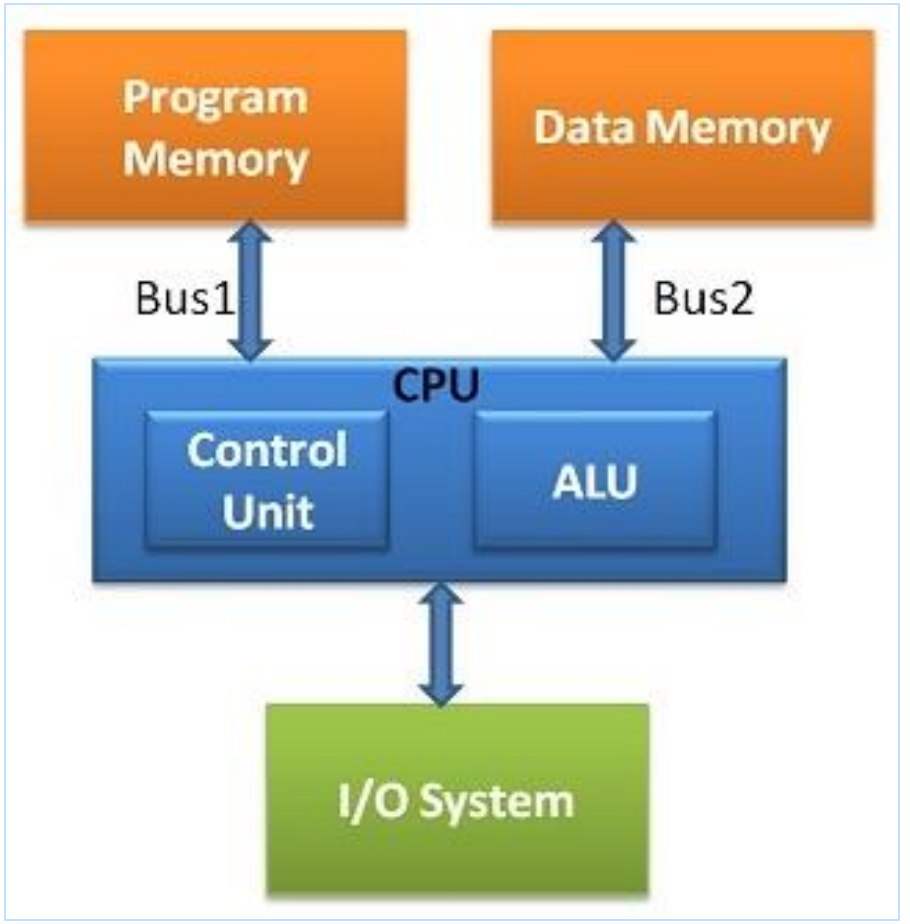
Source: https://witscad.com/course/computer-architecture/chapter/fundamentals-of-architectural-design

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Architecture models (2/3)

- Uses two memory cores to prevent bottleneck.

Harvard architecture



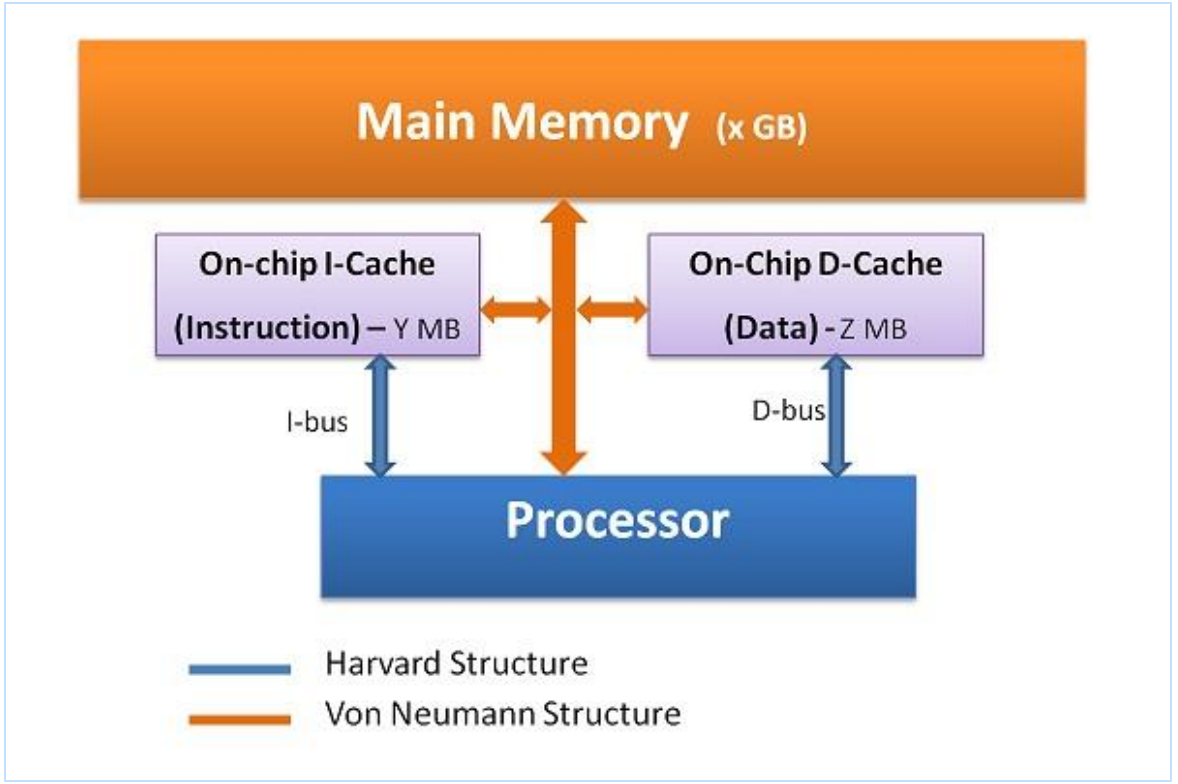
Source: https://witscad.com/course/computer-architecture/chapter/fundamentals-of-architectural-design

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Architecture models (3/3)

- Splits controls into data and instructions.

Merged architecture



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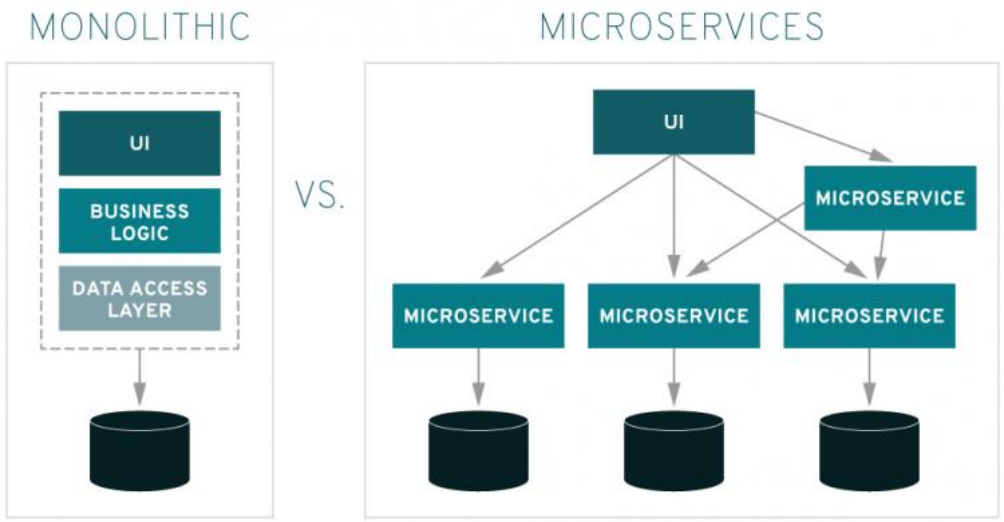
Monoliths and microservices

Monoliths

- Legacy style applications.
- One path to rule them all.
- **Pros:** easy development, performance.
- **Cons:** tight coupling, slow releases.

Microservices

- Build large apps as smaller components.
- Communicate via API.
- Allow decentralized development.
- **Pros:** rapid response to production, scalability, failure isolation.
- **Cons:** development, refactoring, request efficiency.



Source: https://pretius.com/why-you-should-choose-the-microservices-architecture/

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Building from the bottom (1/2)

Evaluate current needs

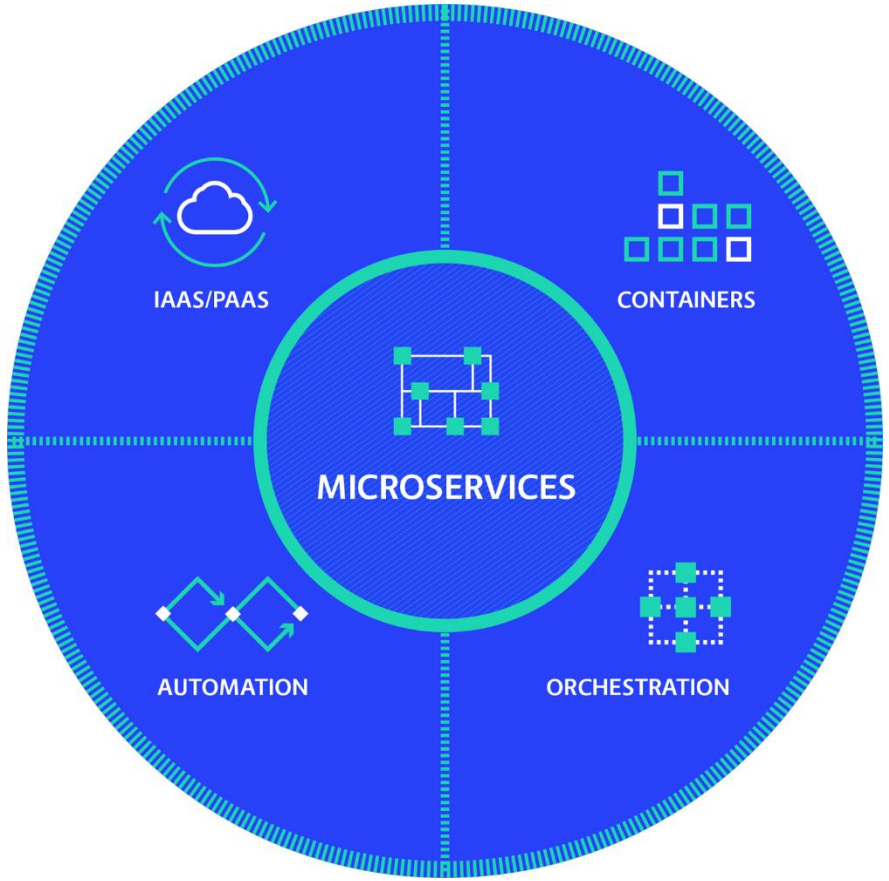
- on-premises
- cloud-using
- cloud-native
- hybrid.

Migrating architecture

- on-premises to cloud native
- monolith to microservices
- cloud-native to cloud-based.

Strangler pattern

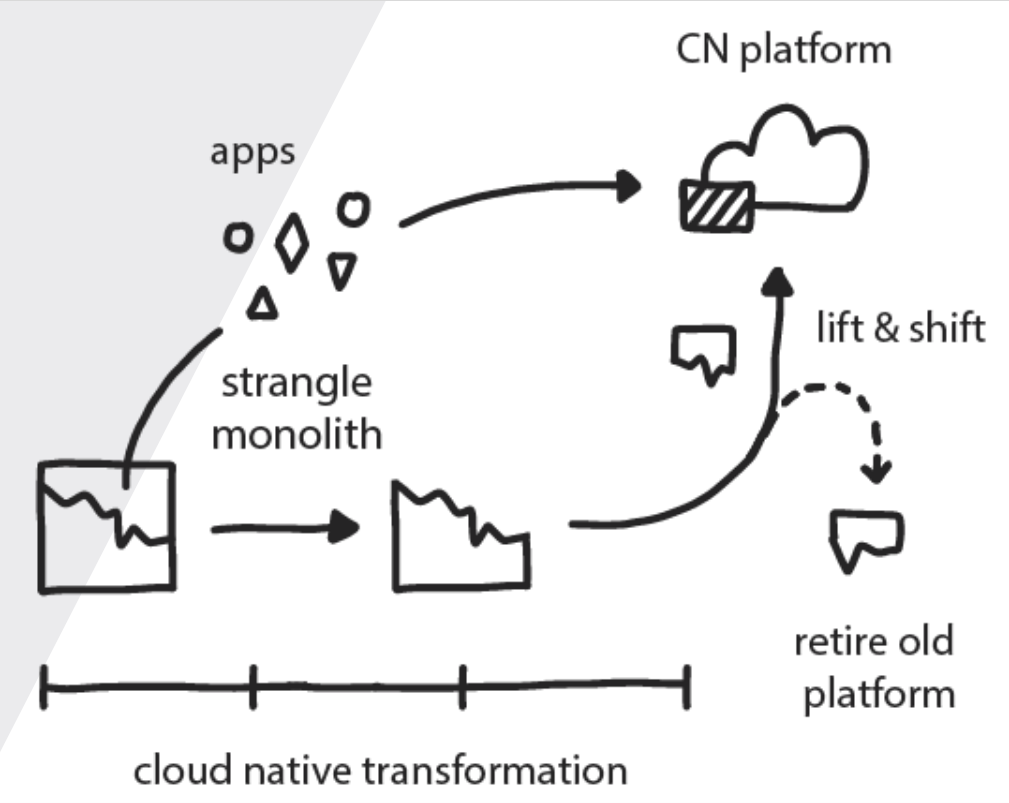
- Apps
- Orgs.



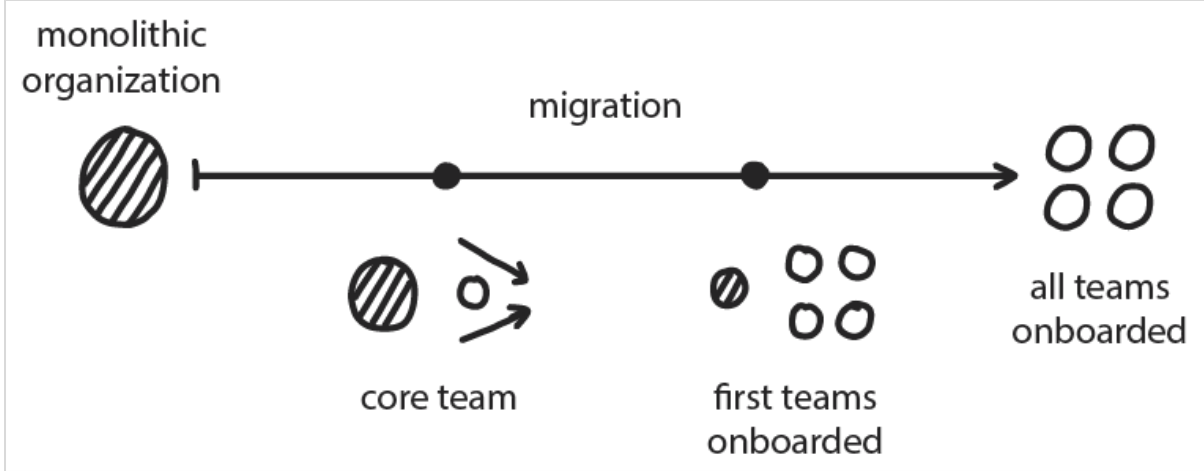
The relationship diagram between the five principles of cloud native architecture
Source: https://www.oreilly.com/library/view/cloud-native-transformation/9781492048893/ch01.html

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Building from the bottom (2/2)



Strangle Monolithic Application
Source: https://www.cnpatterns.org/development-design/strangle-monolithic-application

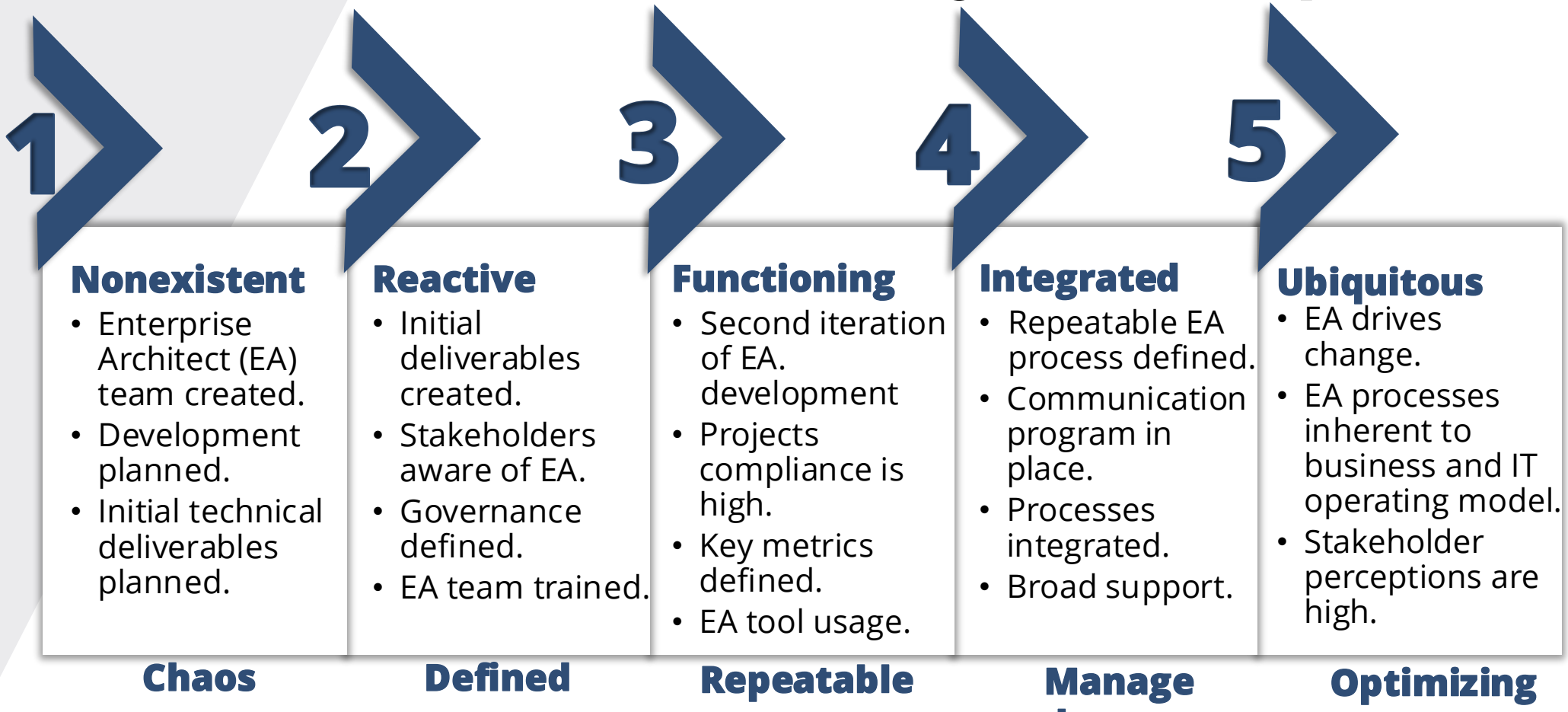


Strangle Monolithic Organization
Source: https://www.cnpatterns.org/organization-culture/strangle-monolithic-organization

For design options in cloudnative:
cnpatterns.org

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Architecture maturity: enterprise



- Maturity guides growth.
- Knowing your requirements supports expansion.

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Architecture maturity: API

API ARCHITECTURE MATURITY MODEL

	Classifications						
	NOT BASED ON APIs			PARTIALLY BASED ON APIs		FULLY BASED ON APIs	
Levels	1	2	3	4	5	6	7
	Isolated Applications	Unstructured Integrations	Component -based Architectures	Service -oriented Architectures	Private APIs based on Microservices	Open APIs	APIs as Business
Main Tech Topics	Monolithic App	Sockets File Sharing Msg Queues DB Replications	EJB RMI COM/COM+ CORBA WebServices	ESB BPEL CEP BAM SOAP	REST AMQP API Gateway Containers	API Platform Dev Portal WAF REST OAS	API Platform Service Mesh Cloud GraphQL gRPC

Source: https://www.apiscene.io/lifecycle/7-layers-of-api-architecture-maturity/

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